Fashion Knit Garments Ltd
H.R. Textile Mills Ltd

4, Karnapara, Savar, Dhaka
(+23.825375N, 90.25754E)

11 May 2014



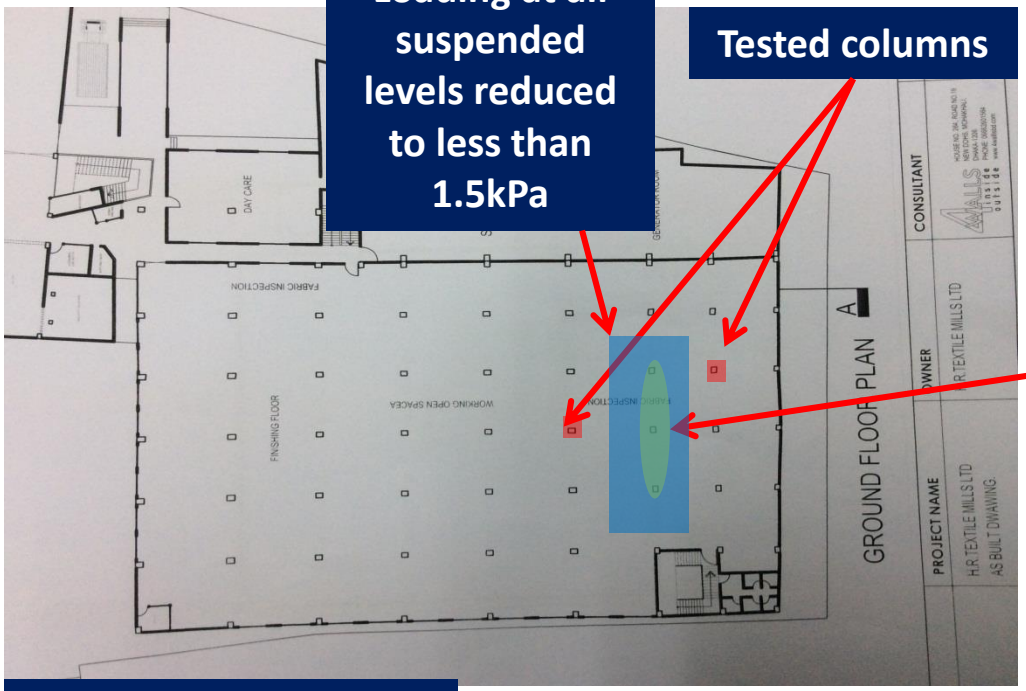
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Observations

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Columns appear to be stressed to a high level and require immediate review



Typical column layout

Columns in this area appear smaller than shown on the drawings – specific check on these columns to be carried out by Building Engineer



Tested Ground Floor Column – Stone Chips

Outline calculations indicate that columns are stressed to a high level and require immediate review.

Building Engineer to perform detailed calculations including a Detail Engineering Assessment (see attached scope) and concrete tests to prove adequacy of column sizes, and (if required):

- Reduce loads by vacating floors
- Reinforce columns

Column Strength

Management of floor storage loads and structural adequacy of slab to support floor build-up loads in toilet areas

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Floor build-up loads in Toilet Area



Grey Fabric store loads

Building Engineer to prepare controlled loading plans, based on floor slab and column capacity, for all floors which will designate allowable storage density and where storage may be placed.

Management of loads/floor build-up

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Mezzanine slab and support columns

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- Cracking to Mezzanine slabs
- Mezzanine slabs supported on columns which are in turn supported on slabs/beams between main gridlines
- Building Engineer to verify adequacy of this arrangement



- Mezzanine support Columns appear to be supported on suspended slab and in some cases beams below.

Mezzanine Area highlighted

Loading/ Mezzanine Support

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Check on beam design required for suspended rack/trolley system

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New racking system erected in 2007 supported from secondary beams

Building Engineer to check beams for point loads from racks and strengthen/reduce load if necessary

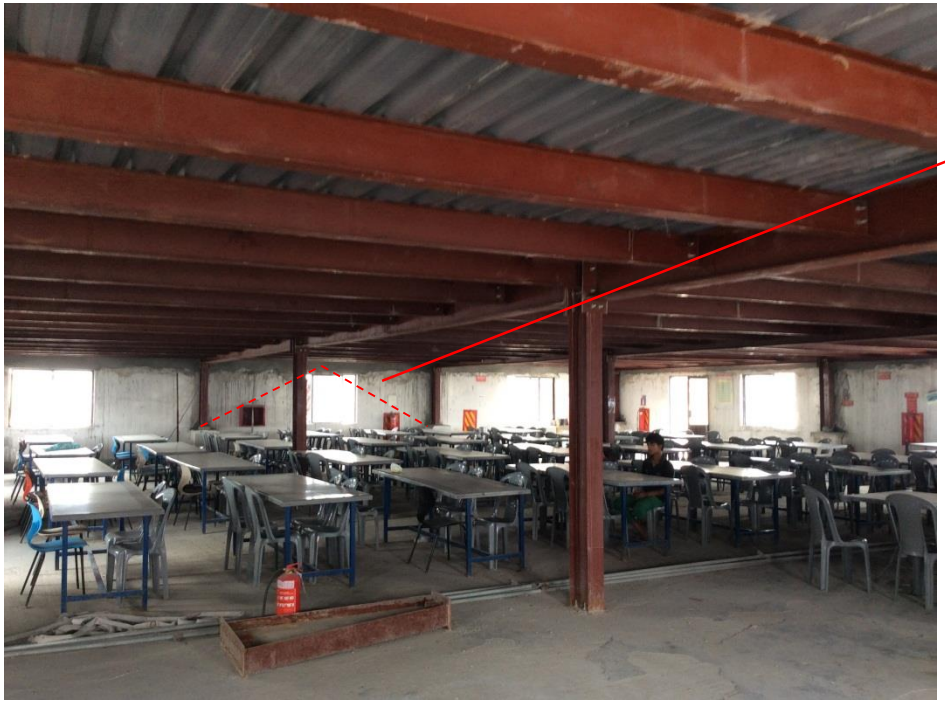


Check on beam design required for suspended rack/trolley system

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Lateral stability of steel frame to roof Dining Area

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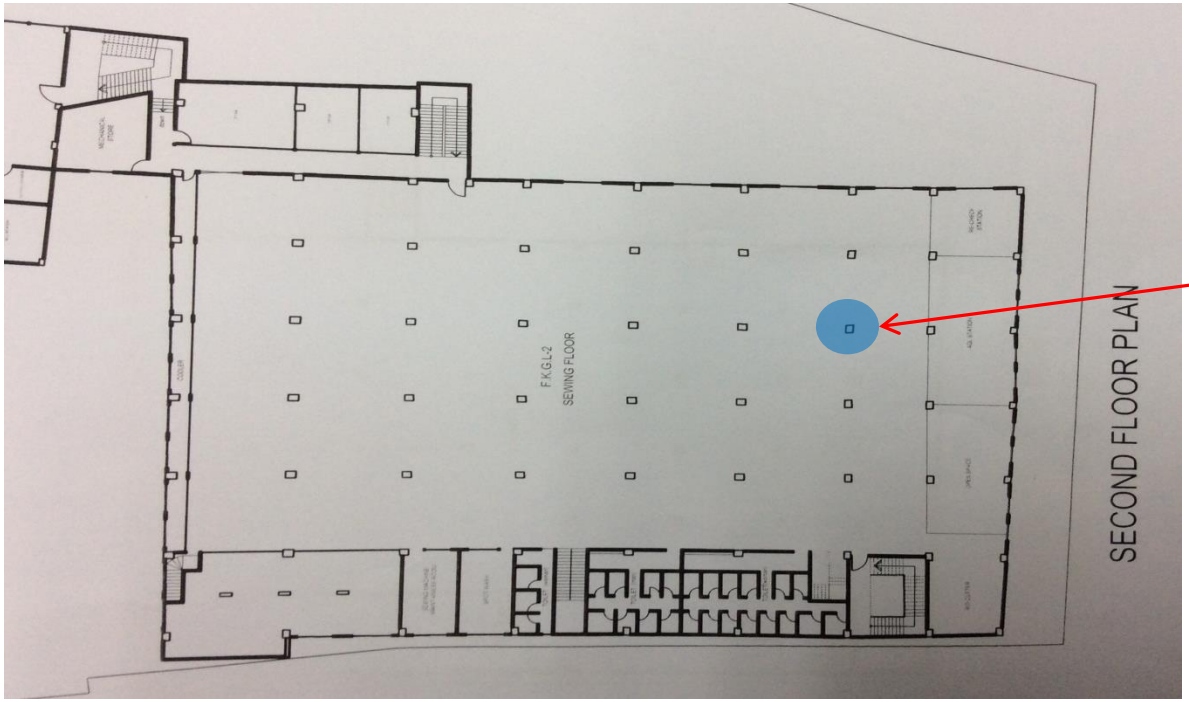
No vertical bracing to steel structure. Building Engineer to check adequacy of steel structure.

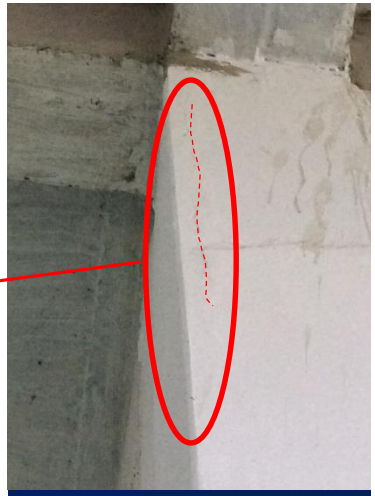
Dining Steel Structure

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Check on column crack/damage in one location

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Vertical crack in column/column render

Building Engineer to investigate column, to establish if cracking is in render only or in concrete and propose repair or strengthening as necessary

Cracking in column

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Check anchorage of communication masts at roof level

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The anchorage of two large communication masts is to be checked and verified by the Building Engineer

Anchorage of communication masts

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Water causing damage and corrosion

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Water causing damage and corrosion

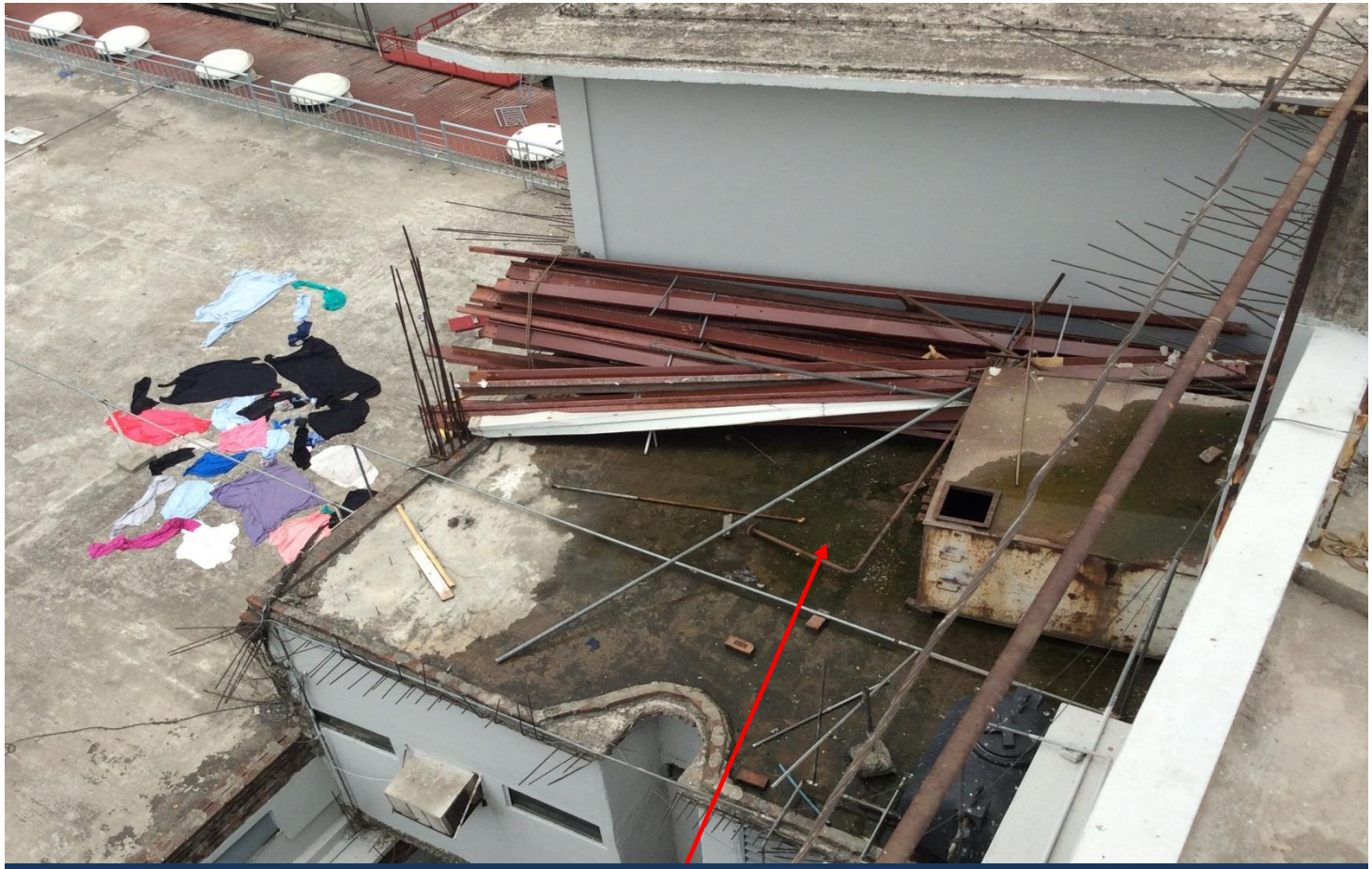
View from Building 1 roof looking at Building 2 ;

Unmanaged roof drainage causing corrosion to façade and structure

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Remove steel stored on roof of Building 1 /Building 2 link

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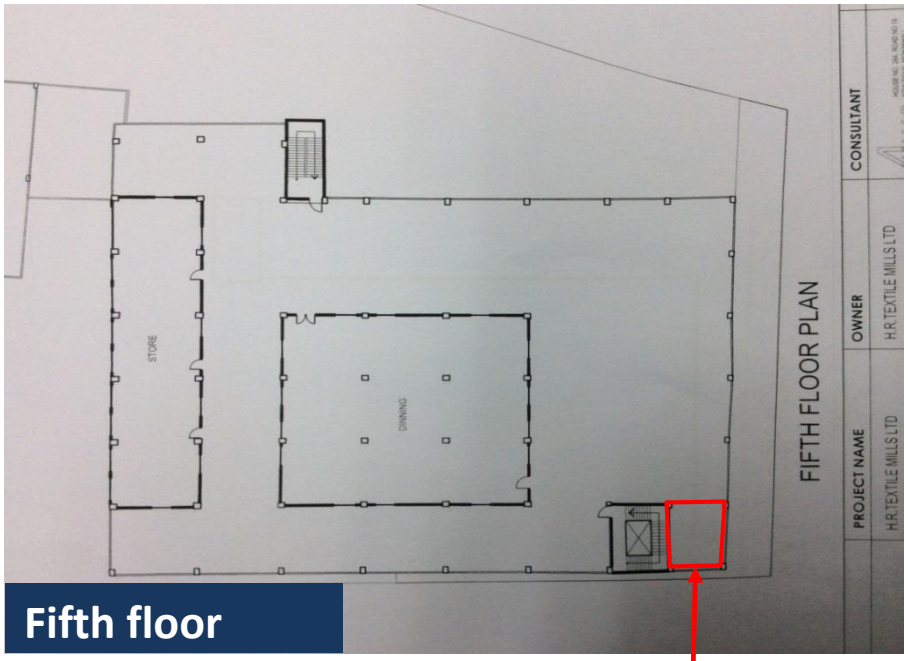
All steel stored at the roof of the structure linking Buildings 1 and 2 to be removed

Steel stored on roof of Building1/Building 2 link

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Discrepancies between drawings provided and the as-constructed building

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Fifth floor

Room built beside staircase

Available drawings do not fully correspond with the as constructed building.

Building Engineer to survey as-built structure and update drawings including design verification

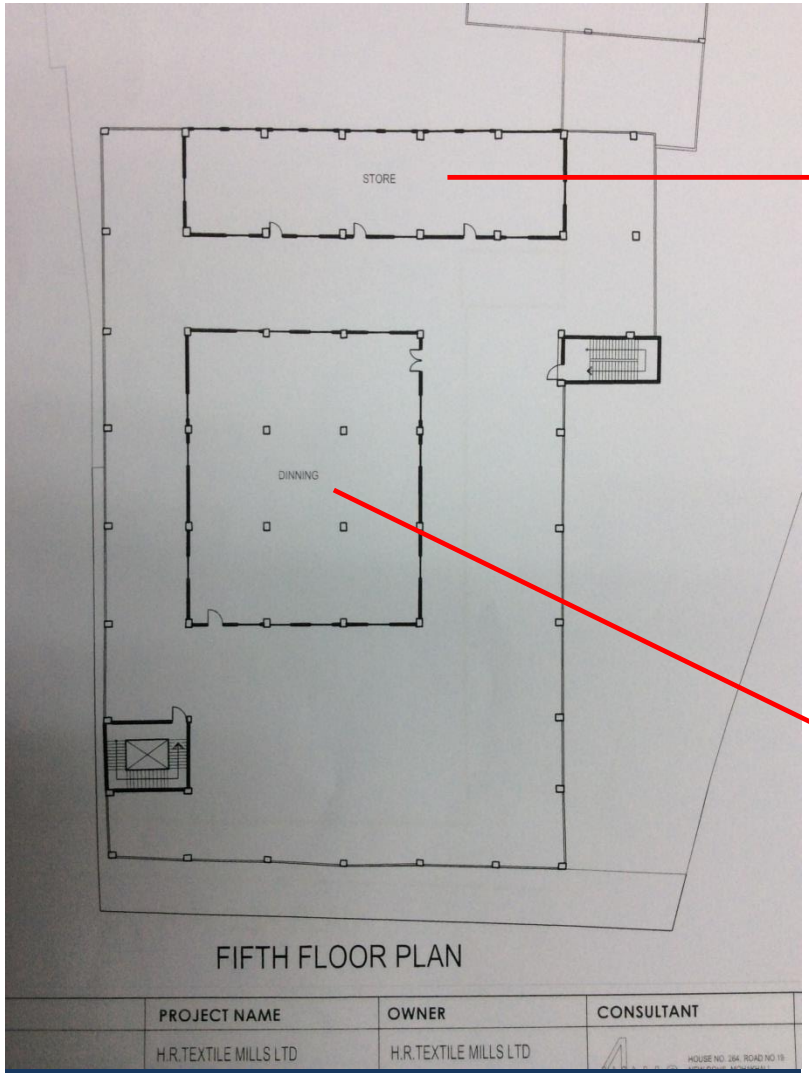
No structural information available for building columns, floor slabs or steel roof construction.

Drawing Discrepancies

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Building 2 Non-Structural Observation
No handrails provided on accessible roofs

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Handrail recommended to Store and Dining area roofs which are accessible



Vacant Rooms & Store



Dining Area

Installation of handrails to roof perimeter is recommended.

No handrails provided on accessible roofs

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Observations on Building 3 – Office Building (not inspected)

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- Water tank on cantilever – structural check should be carried out



- Overall building stability should be verified
- Unrestrained masonry panels should be checked

Observations on Office Building

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Priority Actions

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Problems Observed

ITEM 1: Columns appear to be stressed to a high level and require immediate review

ITEM 2: Management of floor storage loads and structural adequacy of slab to support floor build-up loads in toilet areas

ITEM 3: Mezzanine slab and support columns

ITEM 4: Check beam design for suspended rack/trolley system

ITEM 5: Lateral stability of steel frame to roof Dining Area

ITEM 6: Check on column crack/damage in one location – see photos

ITEM 7: Check anchorage of communication masts at roof level

ITEM 8: Water causing damage and corrosion

ITEM 9: Remove steel stored on roof of Building 1/Building 2 link

ITEM 10: Discrepancies between structural details provided and the as-constructed building

Item No.	Observation	Recommended Action Plan	Recommended Timeline
1	Columns appear to be stressed to a high level and require immediate review	Factory Engineer to review design, loads and column stresses in all columns. Particular check to be completed for columns which are smaller than shown on the drawings – see the full report	**Immediate - Now**
2	Columns appear to be stressed to a high level and require immediate review	Loads to be maintained/reduced to 1.5kPa in area highlighted at all suspended levels	**Immediate - Now**
3	Columns appear to be stressed to a high level and require immediate review	Verify insitu concrete strength either by 100mm diameter cores or existing cylinder strength data for cores from 4 columns . Verify grade of steel reinforcement used.	**Immediate - Now**
4	Columns appear to be stressed to a high level and require immediate review	A Detail Engineering Assessment of Factory to be commenced, see attached Scope.	**Immediate - Now**
5	Columns appear to be stressed to a high level and require immediate review	Detail Engineering Assessment to be completed.	**6-weeks**
6	Columns appear to be stressed to a high level and require immediate review	Produce and actively manage a loading plan for all floor plates within the factory giving consideration to floor capacity and column capacity.	**6-weeks**
7	Columns appear to be stressed to a high level and require immediate review	Continue to implement load plan.	**6-months**

Detail Engineering Assessment

This Schedule develops a minimum level of information, Analysis and testing expected as part of a Detail Engineering Assessment.
The Building(s) have been visually assessed and it is deemed necessary that a detailed engineering assessment be carried out by a competent Engineering Team employed by the factory Owner.
This Request should be read in conjunction with the BUET developed Tripartite Guideline document for Assessment of Structural Integrity of Existing RMG Factory Buildings in Bangladesh (Tripartite Document), the latest version of this document should be referenced. T his document also gives guidance on required competency of Engineering Team.

We expect that the following will be carried out:

1. Development of Full Engineering As-Built Drawings showing Structure, loading, elements, dimensions , levels, foundations and framing on Plan, Section and Elevational drawings .
2. The Engineering team are to carry out supporting calculations with a model based design check to assess the safety and serviceability of the building against loading as set out in BNBC-2006, Lower rate provisions can be applied in accordance with the Tripartite Guidelines following international engineering practice, justification for these lower rate provisions must be made.
3. A geotechnical Report describing ground conditions and commenting on foundation systems used/proposed.
4. A report on Engineering tests carried out to justify material strengths and reinforcement content in all key elements studied.
5. Detailed load plans shall be prepared for each level showing current and potential future loading with all key equipment items shown with associated loads.
6. The Engineering team will prepare an assessment report that covers the following:
    - As-Built drawings including
        - Plans at each level calling up and dimensioning all structural components
        - Cross sectional drawings showing structural beams, slabs, floor to floor heights, roof build-ups and Basic design information of the structure
    - Highlight any variation between As-built compared to the designed structure
    - Results of testing for strength and materials
    - Results of geotechnical assessment and testing/investigation
    - Details of loading, inputs and results of computer modelling
    - Commentary on adequacy/inadequacy of elements of the structure
    - Schedule of any required retrofitting required for safety or performance of Structure

Any proposals for Retrofitting to follow guidance developed in the Tripartite Document

Item No.	Observation	Recommended Action Plan	Recommended Timeline
8	Management of floor storage loads and structural adequacy of slab to support floor build-up loads in toilet areas	Extent of build-up loading within toilet areas to be surveyed and capacity of floor slab to be assessed to confirm that the floor slab is designed to carry these loads.	**6-weeks**
9	Management of floor storage loads and structural adequacy of slab to support floor build-up loads in toilet areas	Produce and actively manage a loading plan for all floor plates within the factory giving consideration to floor capacity and column capacity.	**6-weeks**
10	Management of floor storage loads and structural adequacy of slab to support floor build-up loads in toilet areas	Continue to implement load plan.	**6-months**
11	Mezzanine slab and support columns	Building Engineer to verify structural adequacy of this arrangement – slab, column and transfer beam arrangement. Note that in a number of instances mezzanine columns appear to be supported directly from the floor slab without transfer beams.	**6-weeks**
12	Check beam design for suspended rack/trolley system	Building Engineer to verify structural adequacy of the floor beams to support floor loads and additional loads due to suspended rack/trolley system.	**6-weeks**
13	Lateral stability of steel frame to roof Dining Area.	Building Engineer to verify structural adequacy of this arrangement and design bracing system to be installed if required.	**6-weeks**

Item No.	Observation	Recommended Action Plan	Recommended Timeline
14	Check on column crack/damage in one location – see photos.	Building Engineer to investigate if cracking is superficial only and carry out repairs or strengthening as required.	**6-weeks**
15	Anchorage of communication masts on Building 2 roof	Building Engineer to check and verify anchorage of communication masts on Building 2 roof	**6-weeks**
16	Water causing damage and corrosion	Engineer to inspect water damaged structure including the exterior and propose a suitable repair. Roof drainage system to be upgraded as necessary and new downpipes installed where required.	**6-weeks**
17	Remove steel stored on roof of Building 1/Building 2 link.	Stored structural steel beams to be removed from roof level and stored at ground level.	**6-weeks**
18	Discrepancies between structural details provided and the as-constructed building	Building Engineer to survey as-built structure and prepare accurate as-built survey information, including all columns, beams and slabs as part of DEA.	**6-weeks**